Representation modalities

Domain objects

Family	Object	Sequences	Images	Video	Audio	Graphs	3D geometry	Tabular data	Numerical signals	Spatial fields
Biomolecular objects	DNA	✓	✓	·	·	✓	✓	·		·
	RNA	✓	✓	·	·	✓	✓	·		·
	Proteins	✓	✓	·	·	✓	✓	·		·
	Peptides	✓	✓	·	·	✓	✓	·		·
Chemical and material objects	Small molecules	✓	✓	·	·	✓	✓	·		·
	Chemical reactions	✓	✓	·	·	✓	·	✓		·
	Polymers	✓	✓	·	·	✓	✓	·		·
	Crystals	✓	✓	·	·	✓	✓	·		✓
Biomedical observations	Single-cell expression profiles	✓	·	·	·	·	·	✓	✓	·
	Cell-population expression profiles	✓	·	·	·	·	·	✓	✓	·
	Cell morphologies	·	✓	✓	·	✓	✓	✓	✓	·
	Tissue specimens	✓	✓	·	·	✓	✓	✓	·	✓
	Clinical records	✓	✓	✓	✓	✓	·	✓	✓	·
Physical and engineered objects	PDE solutions	✓	·	·	·	✓	·	·	✓	✓
	Atmospheric states	·	✓	·	·	✓	·	✓	✓	✓
	Geographical regions	✓	✓	✓	·	✓	✓	✓	✓	✓
	Scenes	✓	✓	✓	✓	✓	✓	·	·	✓
	Robot actions	✓	·	·	·	·	✓	✓	✓	·
Digital objects	Language and Programs	✓	·	·	·	✓	·	·	·	·
	Mathematical expressions	✓	✓	·	·	✓	·	·	·	·
	Recommendation items	✓	✓	✓	·	✓	·	✓	✓	·
	User interaction histories	✓	·	·	·	✓	·	✓	✓	·

Biomolecular Complexes

Celluar Systems

World models

Multimodal large language models

User behavior system